

RAJ KUMAR MISHRA

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PROFESSIONAL SUMMARY

Final-year B.Tech CSE (AI/ML) student specializing in backend and full-stack development using *Node.js*, *Express.js*, *React.js*, *Next.js*, *MongoDB*, and *TypeScript*. Experienced in building scalable applications with REST API architecture, authentication systems, AI integration, and performance optimization.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS

Backend: Node.js, Express.js, REST APIs, JWT Authentication, Middleware, MVC Architecture

Databases: MongoDB, MySQL, Supabase

AI & ML: PyTorch, HuggingFace, mBERT, Librosa, scikit-learn

Tools & Concepts: Git, GitHub, Postman, Chrome DevTools, Lighthouse, OOP, DSA

PROJECTS

Multimodal Song Valence Prediction System – NITRA Technical Campus

Developed a multimodal AI system for predicting emotional valence in Hindi and English songs using audio and text fusion models.

- Built a human-annotated Bollywood valence dataset containing 235 Hindi songs.
- Designed a three-stream neural architecture combining mBERT embeddings and Librosa audio features.
- Achieved Pearson correlation scores of 0.53 on English and 0.45 on Hindi datasets.
- Deployed the full-stack application using FastAPI, Next.js, Hugging Face, Vercel, and Supabase.

Technologies Used: PyTorch, HuggingFace, mBERT, Librosa, FastAPI, Next.js, Supabase

BetterMind – AI Resume Analyzer

Developed an AI-powered resume analysis platform focused on scalable backend processing and performance optimization.

- Designed backend APIs for resume processing, authentication, and modular workflows.
- Improved Lighthouse performance score from 65 to 92 through optimization techniques.
- Reduced Largest Contentful Paint (LCP) by 1.2 seconds using lazy loading and rendering optimization.

Technologies Used: Node.js, React.js, TypeScript, MongoDB, Lighthouse

LangChat – Real-Time Chat Platform

Built a scalable real-time chat platform with secure authentication and optimized backend architecture.

- Developed complete backend architecture using Node.js and Express.js with modular MVC structure.
- Designed and implemented 15+ REST API endpoints for authentication, messaging, chat rooms, and profiles.

Technologies Used: Node.js, Express.js, MongoDB, React.js, JWT, Tailwind CSS

ACHIEVEMENTS

- Team Lead – Smart India Hackathon 2024 (Software Edition) Finalist.
- Built and deployed 15+ full-stack applications demonstrating scalable architecture principles.

EDUCATION

B.Tech in Computer Science and Engineering (AI/ML)

Dr. APJ Abdul Kalam Technical University

2022 – 2026 Uttar Pradesh, India

CERTIFICATIONS

Web Development With React

Certificate in Web Development with React from Anudip Foundation

Project Engineer – 5G Network

Certification as Project Engineer in 5G Network from Ericsson

LANGUAGES AND ADDITIONAL INFORMATION

English – Professional | **Hindi** – Native | **Hobbies** – Chess, Sudoku, Reading Technology Blogs